

How Incomplete Exotic Species Management Can Make Matters Worse: Experiments in Forest Restoration in Ontario, Canada

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ABSTRACT

Funding for ecological restoration efforts sometimes ends prematurely and exacerbates problems that originally were the rationale for the project. In our test of methods to manage exotic species, we mimicked this real world situation and compared management after one year with no ecological restoration with exotic species management continued yearly for five years and followed by ecological restoration (transplanting adult plants of four native species) in year five. Our study site in Ontario, Canada consisted of eight fragmented woodlots dominated by sugar maple (*Acer saccharum*) and American beech (*Fagus grandifolia*) that supported dense populations of the exotic species garlic mustard (*Alliaria petiolata*), dame's rocket (*Hesperis matronalis*), and celandine (*Chelidonium majus*). In 1998, we quantified the impact and duration of one-time use of glyphosate, hand-pulling, inflorescence clipping, and mulch on the seedbank and shoots of three exotic species, native herbaceous species, and (for shoots, non-adult) native species of shrubs and trees. By 2006, above- and belowground densities of exotic species significantly increased and native species significantly decreased with one-time treatments of glyphosate and hand-pulling (ANOVA). The exotic species recovered quickly from one-time application of glyphosate, and hand-pulling disturbed the soil and increased relative emergence of exotic species from seed. As expected, inflorescence clipping and mulching were ineffective as one-time treatments but did not worsen the exotic species problem or decrease native species' densities. The repeated management plus ecological restoration was more successful in all cases, with similar results by 2006, while one-time application of glyphosate and hand-pulling worsened problems. We caution that initiating exotic species management can be risky if funding is not secure enough to support longer term efforts.

Keywords: *Alliaria petiolata*, *Chelidonium majus*, ecological restoration project management, exotic species management, *Hesperis matronalis*, Ontario

Exotic species management and restoration attempts often suffer from a lack of systematic funding and political support, and, as a result, efforts can range from token attempts to more viable projects that get either delayed or cancelled (Quon et al. 1999, 2000, Meloche and Murphy 2006). When a project is truncated, exotic species infestations may worsen, and well-intentioned restoration efforts may exacerbate the problem. Not only can resources be wasted and participants disappointed but also

the outcome can be worse than doing nothing. We decided to test outcomes of managing exotic species for one year with no attempt to reestablish native vegetation against continuous management followed by replanting. We chose this comparison because actual projects often begin with ambitious plans to manage invasive species and restore native species, but funding does not sustain either effort. Our study tested the alternatives using a controlled experimental design.

We focused on three well-studied exotic species (Murphy 2005): garlic mustard (*Alliaria petiolata*), dame's rocket (*Hesperis matronalis*), and celandine, (*Chelidonium majus*). All three species colonize fragmented woodlands

and all tend to outcompete understory herbs and seedlings of shrubs and trees (Cavers et al. 1979, Haber 1996, 1998, Stinson et al. 2006). Garlic mustard is an adaptable and plastic species that varies in response to management (Nuzzo 1991, 1994, Carlson and Gorchov 2004). It is a prolific seed producer (Susko and Lovett Doust 1998, 1999, 2000) that is allelopathic (McCarthy and Hanson 1998), competitive for light and nutrients (Meekins and McCarthy 1999), and harmful to arbuscular mycorrhizal fungi (Roberts and Anderson 2001, Stinson et al. 2006). Dame's rocket and celandine are not as well studied. In northeastern North America, they often are sympatric with garlic mustard,

although dame's rocket tends to colonize sunnier, mesic areas (Mitchell and Ankeny 2001, Rothfels et al. 2002) and allelopathic celandine colonizes shadier, dry areas (Tome and Colombo 1995, Gorb and Gorb 2000). Slight niche separation could allow all three species to spread into woodland fragments with no competitive interaction occurring among them.

We quantified the outcome and duration of effect of a single management effort in 1998 consisting of a glyphosate application, hand-pulling, inflorescence clipping, and mulching on the seedbank and shoots of three exotic species, native herbaceous species, and juveniles (shrubs and saplings) of native species of shrubs and trees. We compared this single effort to repeated exotic species management efforts between 1998 and 2002 along with ecological restoration using three- to five-year-old shoots grown from rescued stock that were no more than 2 km away from the transplant site. The species used were trout lily (*Erythronium americanum*), sharp-lobed hepatica (*Hepatica acutiloba*), mayapple (*Podophyllum peltatum*), and bloodroot (*Sanguinaria canadensis*).

Study Site and Experimental Design

In 1998, we selected two sites each in the Regional Municipality of Waterloo, the Regional Municipality of Hamilton, Wellington County and Oxford County in Ontario, Canada. In Ontario, the political administrative terms "regional municipality" and "county" denote government units of approximately the same jurisdiction. All sites were fragmented woodlots dominated locally by sugar maple (*Acer saccharum*) and American beech (*Fagus grandifolia*). The sites were spread along a 120-km arc from Hamilton (in the south) to Wellington County (in the north). All except the Hamilton sites were on private land; Hamilton sites were located on land administered by the Hamilton Region Conservation Authority, a provincial government

agency operating to protect terrestrial and aquatic resources at a landscape scale. All locations had similar mean seedling densities and population densities of the three exotic species and similar total populations of native herbs and seedlings of shrubs and trees. These similar initial densities allowed us to test for differences in treatment effects related to local weather. All study sites were within five meters of trail corridors and were located at the edge of forest tracts along trails that were generally devoid of large mature trees or shrubs. Blocks, plots, and replicates represented influences of microclimate on treatment effectiveness.

At each of the three study sites we tested the response of exotic and native species to one of four common treatments for exotic species management. Treatments were applied using a design with 12 replicate plots completely randomized block per treatment. Treatment plots were 1 m × 1 m. To avoid effects of treatments between neighboring plots, we planted 2-m wide guard rows that separated plots from one another and from surrounding vegetation. In addition to a control, we randomly assigned one of the following four treatments to exotic species plus a control:

- Applied glyphosate Monsanto's Roundup® - 0.9 kg active ingredient/ha) to the whole plot in early spring (not spot applications, because the infestations were dense and seedlings were numerous).
- Hand-pulled all individuals of the three exotic species as soon as stems bolted (typically mid May through mid June).
- Clipped the inflorescences by hand as the flowers begin to senesce (usually late May through late June); the clippings were bagged and removed off site.
- Applied a 5-cm layer of mulch comprised of chipped eastern red cedar (*Juniperus virginiana*) in early spring.

In eight of the plots we applied the treatment only once, in 1998. In the other eight plots, the treatments

were repeated each spring from 1998 through 2002. Also in 2002, we transplanted 2 shoots/m² of trout lily, sharp-lobed hepatica, mayapple, and bloodroot. No further treatments were applied to any plots after 2002.

We assessed plant densities the day before treatment and the day after. In 1998–2002, we counted plants weekly, although we report the densities near the time of treatment. Plots were monitored through fall 2006. Before treatments, we counted the individuals of all species within four 10-cm × 10-cm quadrats placed randomly within each 1-m × 1-m plot. Before treatment each year, we also collected 100 soil cores at 5 cm depth in each quadrat for seedbank analysis. We used a soil auger to collect a 188-cm³ cylindrical soil core. Soil sampling was performed randomly along a W-shaped transect each April. After six months of cold stratification, cores were mixed with peat moss and placed in flats in a growth room with a 12:12 photoperiod with ambient light at 950 micromoles/m²/s photosynthetic photon flux density. These growth conditions represent the conditions found in the regions during early April when many species germinate. Seedling counts were cumulative, that is, the cores collected in 1998 were counted through 2006. This count allowed us to observe germination in native species with complex dormancy and emergence requirements.

Analysis and Results

To identify effects of management treatments on the shoot densities of the three exotic and the native species, we compared impacts of timing and type of treatment with a multivariate repeated measures analysis of variance (MANOVA) (Scheiner 1993, Von Ende 1993). Contrasts with a Bonferroni correction ($p = 0.05$ divided by n comparisons) were used to compare effects of treatments within years and within treatments between years. Similar tests were used for seed germination data.

Table 1. MANOVAR testing for response to management treatments in the numbers of shoots per m². For between-subjects effects, analyses are reported as the standard used for repeated-measured ANOVA (MS, F, P). For within subjects effects, analyses are reported using F, P, and Pillai's Trace. NS: Non-significant at p = 0.05; *: p < 0.05; **: p < 0.01; *: p < 0.001. Interactions that were not significant are not reported below.**

		garlic mustard	dame's rocket	celandine	native herbaceous species	native shrub & tree seedlings
Region or County	MS	1.25	1.02	1.17	1.11	1.31
	F	0.42	0.35	0.46	0.37	0.48
	P	NS	NS	NS	NS	NS
Site	MS	2.37	2.41	2.75	3.04	2.92
	F	0.78	0.80	1.01	1.23	1.14
	P	NS	NS	NS	NS	NS
Block in site	MS	2.31	1.97	2.91	2.47	2.02
	F	0.71	0.48	1.00	0.83	0.61
	P	NS	NS	NS	NS	NS
Treatment	MS	39.89	41.09	39.65	51.89	61.89
	F	27.14	32.52	26.84	47.56	52.37
	P	***	***	***	***	***
Time	Pillai	0.91	0.89	0.84	0.86	0.57
	F	69.71	65.45	62.43	64.13	35.79
	P	***	***	***	***	***
Time x Treatment	Pillai	0.93	0.87	0.77	0.62	0.51
	F	71.02	65.41	59.90	37.76	32.82
	P	***	***	***	***	***

Table 2. Within-treatment contrasts (with a Bonferroni correction) between 1998 pretreatment and 2006 post-treatment plant densities (shoots/m²). "Pretreatment" refers to densities the day before any treatments were applied in 1998. Only the three exotic species experienced hand-pulling and clipping. Single refers to one treatment in 1998. Extended refers to treatments repeated from 1998 to 2002, followed by restoration plantings. Significant differences are due to increases in exotic species in single treatments and decreases in extended treatments. NS: Non-significant at p = 0.05; *: p < 0.05; **: p < 0.01; *: p < 0.001.**

	garlic mustard		dame's rocket		celandine		native herbaceous species		native shrub & tree seedlings	
<i>Pretreatment in 1998 vs. 2006 for:</i>	F	P	F	P	F	P	F	P	F	P
Control	24.32	***	15.67	***	18.61	***	10.18	**	5.41	*
<i>Single Treatment</i>										
Glyphosate	42.11	***	25.12	***	17.67	***	19.12	***	6.35	*
Hand pulled	48.27	***	27.43	***	21.24	***	17.65	***	7.14	*
Clipped inflorescences	25.15	***	16.09	***	18.16	***	10.93	***	5.16	*
5 cm mulch	7.03	*	6.89	*	8.17	*	6.74	*	2.27	NS
<i>Extended Treatments</i>										
Glyphosate	34.91	***	39.08	***	21.13	***	17.67	***	11.21	**
Hand pulled	21.28	***	22.74	***	16.54	***	14.32	***	10.45	**
Clipped inflorescences	24.53	***	21.52	***	24.57	***	27.51	***	15.77	***
5 cm mulch	40.13	***	41.45	***	35.78	***	39.04	***	18.90	***

Table 3. MANOVA testing for response to management treatments in the numbers of seeds per m³. For between-subjects effects, analyses are reported as the standard used for repeated-measured ANOVA (MS, F, P). For within subjects effects, analyses are reported using F, P, and Pillai's Trace. NS: Non-significant at p = 0.05; *: p < 0.05; **: p < 0.01; ***: p < 0.001. Interactions that were not significant are not reported below.

		garlic mustard	dame's rocket	celandine	native herbaceous species	native shrub & tree seedlings
Region/County	MS	1.01	1.50	1.22	1.87	1.82
	F	0.45	0.60	0.58	0.91	0.87
	P	NS	NS	NS	NS	NS
Site	MS	1.96	1.76	2.14	2.25	2.36
	F	0.94	0.73	1.00	1.12	1.19
	P	NS	NS	NS	NS	NS
Block in site	MS	2.42	1.78	2.65	2.98	2.90
	F	1.53	0.75	1.73	1.87	1.84
	P	NS	NS	NS	NS	NS
Treatment	MS	10.89	14.16	13.35	6.71	6.49
	F	5.79	9.13	8.94	3.86	3.63
	P	**	**	**	**	**
Time	Pillai	0.29	0.38	0.32	0.15	0.13
	F	17.71	22.56	20.42	7.96	7.14
	P	***	***	***	*	*
Time x Treatment	Pillai	0.35	0.47	0.38	0.20	0.17
	F	19.82	38.01	21.85	10.11	9.12
	P	***	***	***	**	**

Table 4. Within-treatment contrasts (with a Bonferroni correction) between 1998 pretreatment and 2006 post-treatment seed densities (seeds/m³). "Pretreatment" refers to densities the day before any treatments were applied in 1998. Only the three exotic species experienced hand-pulling and clipping. Single refers to one treatment in 1998. Extended refers to treatments repeated from 1998 to 2002, followed by restoration plantings. Significant differences are due to increases in exotic species in single treatments and decreases in extended treatments. NS: Non-significant at p = 0.05; *: p < 0.05; **: p < 0.01; ***: p < 0.001.

	garlic mustard		dame's rocket		celandine		native herbaceous species		native shrub & tree seedlings	
<i>Pretreatment in 1998 vs. 2006 for:</i>	F	P	F	P	F	P	F	P	F	P
Control	14.91	***	13.67	***	16.06	***	10.31	***	7.77	**
<i>Single Treatment</i>										
Glyphosate	28.09	***	39.14	***	45.81	***	16.15	***	9.15	**
Hand pulled	27.12	***	45.89	***	52.33	***	19.76	***	9.84	**
Clipped inflorescences	13.46	***	13.90	***	15.47	***	10.24	***	7.36	**
5 cm mulch	11.25	***	12.53	***	13.88	***	10.02	***	6.80	*
<i>Extended Treatments</i>										
Glyphosate	15.43	***	14.22	***	15.91	***	8.54	**	7.52	**
Hand pulled	13.01	***	12.75	***	11.84	***	7.58	**	6.44	**
Clipped inflorescences	22.86	***	18.71	***	17.64	***	11.27	***	8.23	**
5 cm mulch	23.57	***	19.06	***	18.49	***	13.50	***	8.81	**

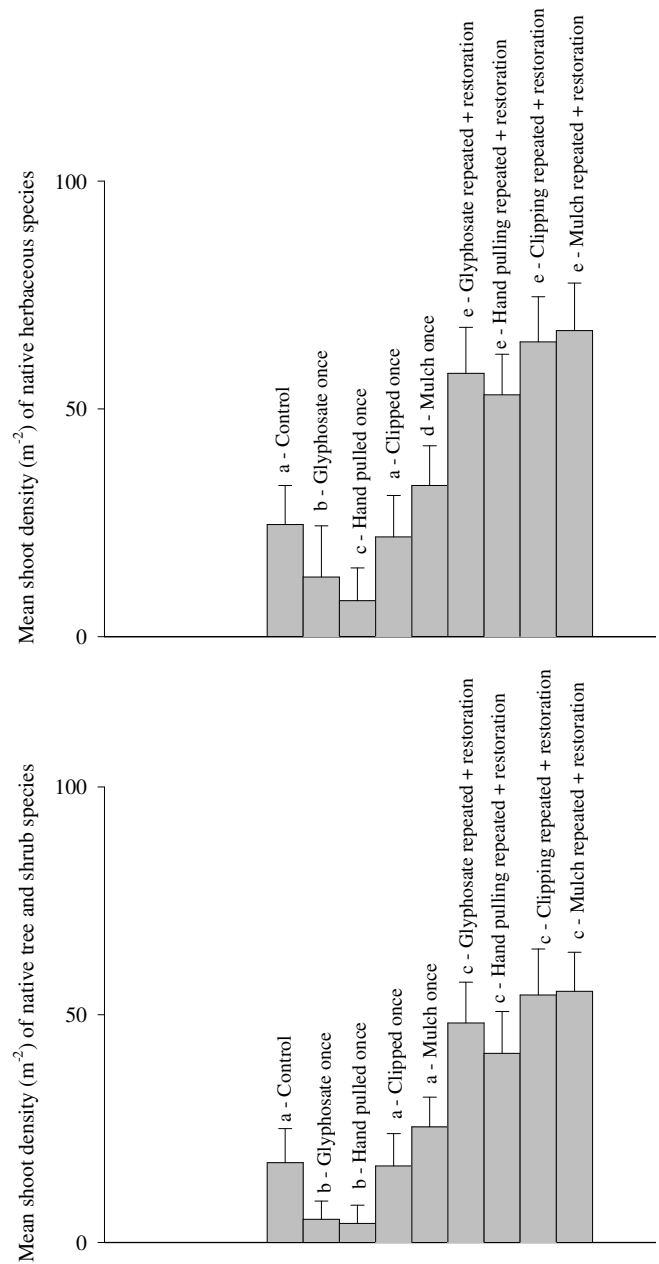
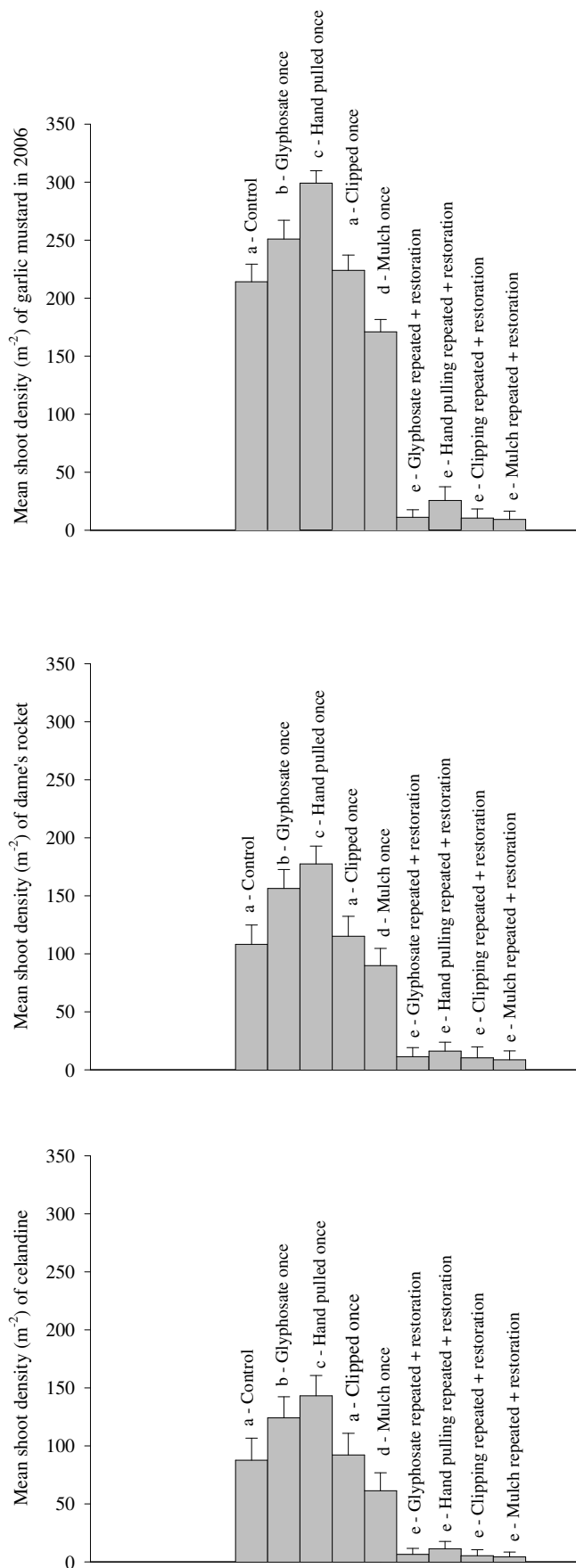


Figure 1. 2006 mean ($\pm$ sd) shoot density in response to treatments from 1998-2002: a) garlic mustard; b) dame's rocket; c) celandine; d) nonnative herbaceous understory species; e) nonnative shrub and tree seedlings. Bars with the same letter are not significantly different from one another according to contrasts run with a Bonferroni correction.

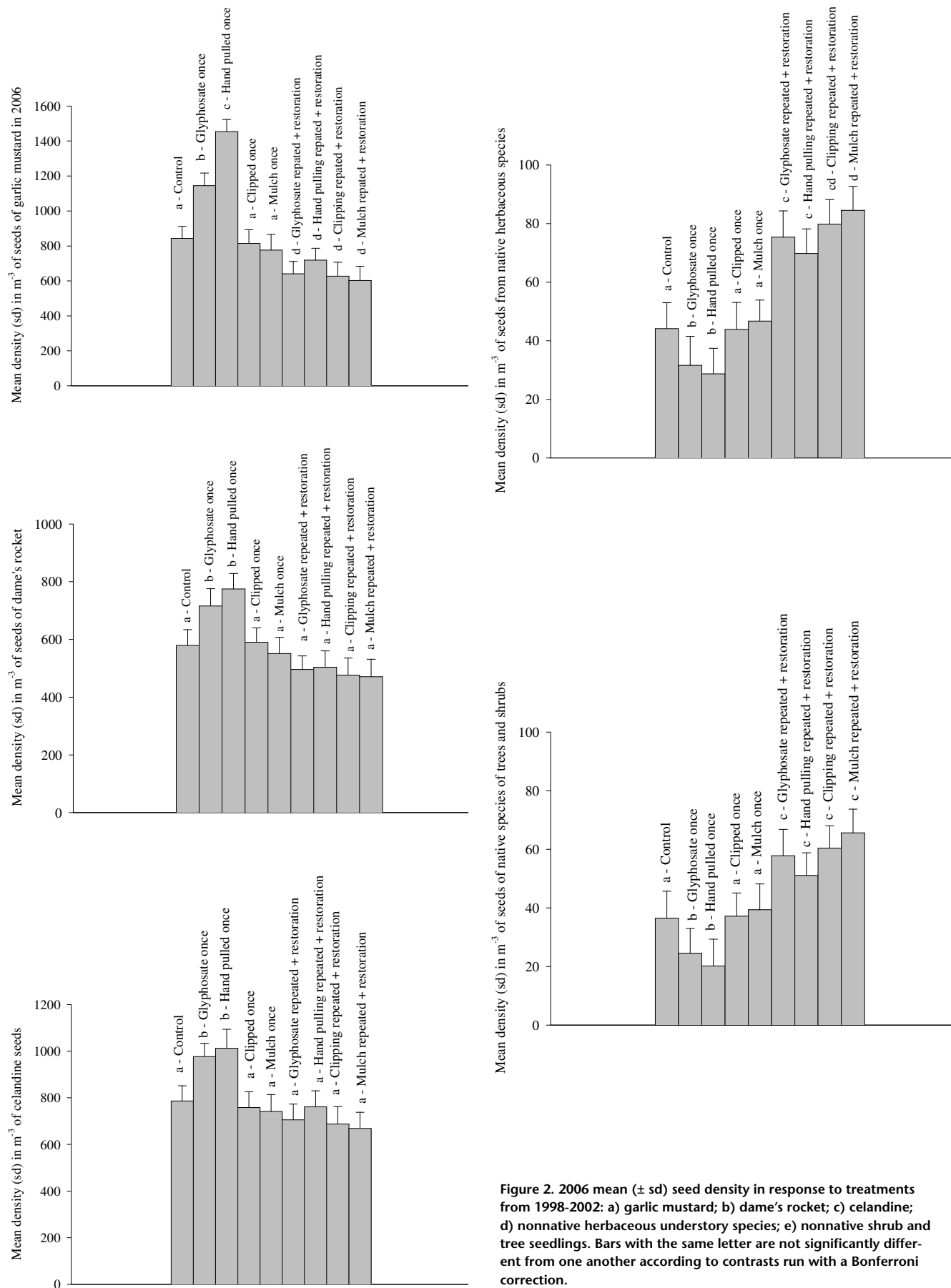


Figure 2. 2006 mean ($\pm$ sd) seed density in response to treatments from 1998-2002: a) garlic mustard; b) dame's rocket; c) celandine; d) nonnative herbaceous understory species; e) nonnative shrub and tree seedlings. Bars with the same letter are not significantly different from one another according to contrasts run with a Bonferroni correction.

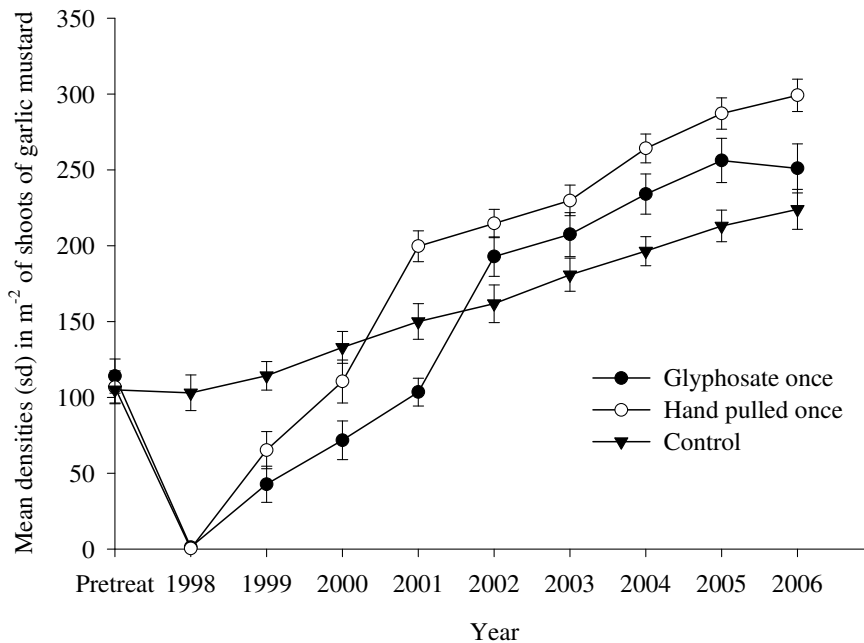


Figure 3. Shoot density of garlic mustard in response to single glyphosate and hand-pulling treatments in 1998 compared to control plot.

We found no significant differences between regions and no interactions of region and treatment or any other spatial relations, allowing us to conclude that the regional population densities did not affect treatment results (Table 1). One-time treatments in 1998 significantly increased densities and seedbanks of the three exotic species and decreased densities and seedbanks of native species by 2006 (Tables 1–4 and Figures 1 and 2). The only exception was that a one-time mulch application did not significantly affect densities of native species of shrubs and trees. By 2006, the one-time glyphosate application and hand-pulling treatment had actually worsened the problem because shoot and seed densities of the three exotic species were significantly higher and densities of native species significantly lower than in the control.

In contrast, in plots where we repeated exotic species treatments between 1998 and 2002 and where we had transplanted native species once in 2002, we significantly reduced densities of the three exotic species and increased densities of the native species relative to pre-treatment conditions and the control. This decrease and increase was true for densities of both

shoots and seedlings that emerged from the seedbank. We chose garlic mustard to show how quickly shoot densities changed. It took about three years for densities in glyphosate and hand-pulled treatments to exceed control plots (Figure 3). While not immediate, the effect was relatively quick, reflecting rapid emergence of seedlings. Native herbaceous species, by comparison, declined in emergence of seedlings and in shoot densities. These results were consistent for dame's rocket, celandine, and native species of trees and shrubs (not shown in detail but see Tables 1–4 and Figure 2).

Implications for Ecological Restoration

Our nine-year study had two main results. First, at any of our sites where we repeated exotic species management for five years and then installed transplants of woodland forbs, we reduced exotics and enhanced native plants. Second, not only did one-time treatment management for exotic species not work (as expected), but glyphosate and hand-pulling treatments actually worsened the problem significantly. This second outcome

has serious implications for restorationists. If budgets cannot support work beyond one year, exotic species management might best be postponed until multi-year funding is assured. Our study challenges the idea that any management is better than nothing.

Glyphosate applications

When glyphosate was applied once and no native species were transplanted, the populations of the three exotic species increased, and native species declined significantly through 2006. Glyphosate is a broad spectrum herbicide with a short residual time. It kills the shoots of all species but has little effect on seed germination. Because our sites had relatively large seedbanks of the three exotic species, germination occurred a week or so post-application, indicating ability to respond under varied environmental conditions. Native species, on the other hand, were constrained by complex dormancy and symbiont requirements (Nuzzo 1999, Luken and Shea 2000, Booth et al. 2003) and perhaps lower seedbank densities, indicated by fewer emerging seedlings. Native species can be constrained by hydrothermal conditions (amount of water interacting with cumulative temperature effects in soil) and photoperiod (Roman et al. 2000). Hence, many native species may not be able to resprout from perennating organs or germinate from seeds until their required conditions occur. By then, it may be too late for them to compete with the dense swaths of exotic seedlings and adults. The biennial life cycle of garlic mustard is a further complication (Winterer et al. 2005). More precise spot application might avoid drift effects of glyphosate on native species but is feasible only where infestations are small enough to allow wick or band applications. Nonetheless, a broadcast herbicide can be effective if natives are transplanted immediately (Murphy 2005). Other herbicides might be less damaging to native species, but none seems selective enough to avoid such damage. The most promising

approach could be a pre-emergence or similar herbicide combined with an early application time. This approach might kill the early germinating garlic mustard and celandine before most native species germinate or become metabolically active. However, non-selective herbicides, like glufosinate, are not as effective against celandine (Jensen et al. 2000). Furthermore, early application would not work on later germinating dame's rocket and the many garlic mustard and celandine seeds that germinate later in the season. Early application also does not address the potential for long-term selection toward later-germinating genotypes that might be selected for by repeated use of herbicides.

Hand-pulling

The likely reason that one-time hand-pulling increased exotic species densities by 2006 is their responsiveness to soil disturbance and the high survival of seedlings to maturity. While some native species might benefit from an immediate release of competition from exotic species or from soil loosening, few native species emerged in response to one-time glyphosate or hand-pulled disturbance (Figure 3a, b). Thus, if there is a dense seedbank of exotic species, one-time and even repeated hand-pulling can exacerbate exotic species densities unless ecological restoration is implemented (Luken et al. 1997, Booth et al. 2003).

Inflorescence clipping

Inflorescence clipping does not disturb the soil, and its impact was similar to control plots. As expected, clipping was only effective when repeated and followed by transplantation of natives. We may have eliminated the "interest" from the seedbank, but there was plenty of "principal" left to ensure exotic species emergence every year. Repeated inflorescence clipping and transplantation might be effective in open areas where a mower could clip tall inflorescences of exotic species but not shorter ones of native species.

Mulching

One-time mulching delayed recovery of exotic species, and repeated mulching was very effective. Mulching significantly reduced shoot numbers of the three exotic species by 2006 and it did not affect native species. Mulching reduces light penetration, challenging exotic species that rely on light for early seed germination, emergence, and shoot growth. Most of the native species are clonal, and ramets are less reliant on seed germination for short-term recruitment and population growth. Their seeds may also be more tolerant of shade, although the mulch layer was thick enough to reduce seed germination. Thus, of the treatments we tested, mulching was the only effective one-time management for garlic mustard, dame's rocket, and celandine during a five-year period. How long mulching can reduce populations of these exotic species if the guard rows exacerbated the effect of the mulch and if native species seed germination was significantly reduced remains to be tested.

Conclusion

In the real world, long-term treatment of exotic species is not always assured, and native species will not necessarily be reintroduced. Our research shows, however, that initiating exotic species management can be risky if funding is not secure enough to support longer-term efforts. In particular, hand-pulling of exotic species and single applications of herbicide proved to be worse than doing nothing.

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